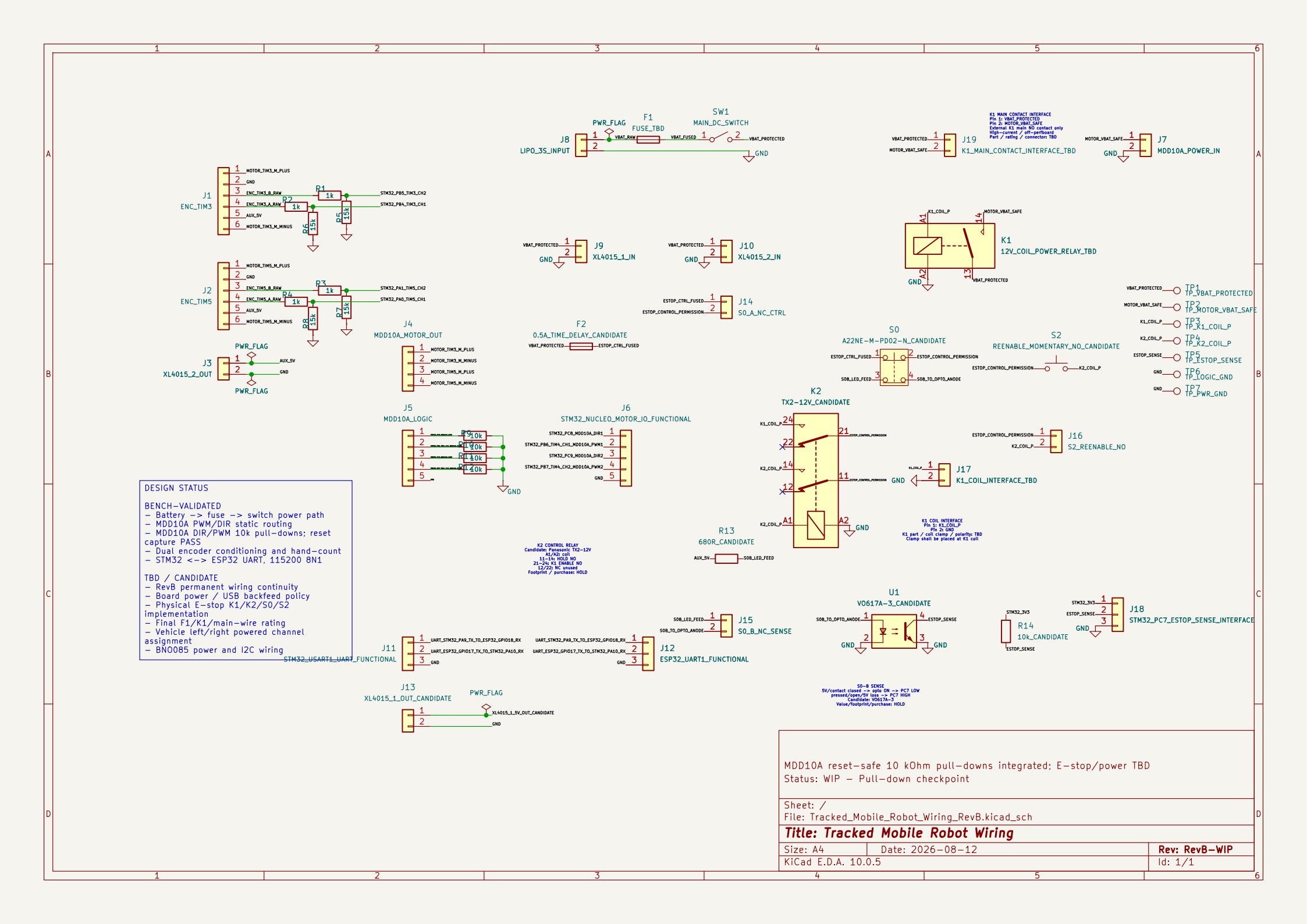

DESIGN STATUS
BENCH-VALIDATED
- Battery -> fuse -> switch power path
- MDD10A PWM/DIR static routing
- MDD10A DIR/PWM 10k pull-downs; reset capture PASS
- Dual encoder conditioning and hand-count
- STM32 <-> ESP32 UART, 115200 8N1
TBD / CANDIDATE
- RevB permanent wiring continuity
- Board power / USB backfeed policy
- Physical E-stop K1/K2/S0/S2 implementation
- Final F1/K1/main-wire rating
- Vehicle left/right powered channel assignment
- BNO085 power and I2C wiring
J1 ENC_TIM3
J2 ENC_TIM5
J3 XL4015_2_OUT
J4 MDD10A_MOTOR_OUT
J5 MDD10A_LOGIC
J6 STM32_NUCLEO_MOTOR_IO_FUNCTIONAL
J7 MDD10A_POWER_IN
J8 LIPO_3S_INPUT
J9 XL4015_1_IN
J10 XL4015_2_IN
J11 STM32_USART1_UART_FUNCTIONAL
J12 ESP32_UART1_FUNCTIONAL
J13 XL4015_1_OUT_CANDIDATE
J14 S0_A_NC_CTRL
J15 S0_B_NC_SENSE
J16 S2_REENABLE_NO
J17 K1_COIL_INTERFACE_TBD
J18 STM32_PC7_ESTOP_SENSE_INTERFACE
J19 K1_MAIN_CONTACT_INTERFACE_TBD
F1 FUSE_TBD
F2 0.5A_TIME_DELAY_CANDIDATE
SW1 MAIN_DC_SWITCH
K1 12V_COIL_POWER_RELAY_TBD
K2 TX2-12V_CANDIDATE
S0 A22NE-M-PD02-N_CANDIDATE
S2 REENABLE_MOMENTARY_NO_CANDIDATE
U1 VO617A-3_CANDIDATE
R13 680R_CANDIDATE
R14 10k_CANDIDATE
TP1 TP_VBAT_PROTECTED
TP2 TP_MOTOR_VBAT_SAFE
TP3 TP_K1_COIL_P
TP4 TP_K2_COIL_P
TP5 TP_ESTOP_SENSE
TP6 TP_LOGIC_GND
TP7 TP_PWR_GND
K1 MAIN CONTACT INTERFACE
Pin 1: VBAT_PROTECTED
Pin 2: MOTOR_VBAT_SAFE
External K1 main NO contact only
High-current / off-perfboard
Part / rating / connectors: TBD
K1 COIL INTERFACE
Pin 1: K1_COIL_P
Pin 2: GND
K1 part / coil clamp / polarity: TBD
Clamp shall be placed at K1 coil
K2 CONTROL RELAY
Candidates: Panasonic TX2-12V
A1/A2: coil
11-14: HOLD NO
21-24: K1_ENABLE NO
12/22: NC unused
Footprint / purchase: HOLD
S0_B SENSE
5V/contact closed -> opto ON -> PC7 LOW
pressed/open/5V loss -> PC7 HIGH
Candidate: VO617A-3
Value/footprint/purchase: HOLD
MDD10A reset-safe 10 kOhm pull-downs integrated; E-stop/power TBD
Status: WIP - Pull-down checkpoint
Sheet: /
File: Tracked_Mobile_Robot_Wiring_RevB.kicad_sch
Title: Tracked Mobile Robot Wiring
Size: A4
Date: 2026-08-12
Rev: RevB-WIP
KiCad E.D.A. 10.0.5
Id: 1/1